Economic Attractiveness and Erasmus Mobility

A descriptive analysis of European student flows

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Text Analysis and Spatial Data for Economists — SP 2026

1. Research Question and Motivation

Erasmus mobility is jointly a spatial-flow problem and a text-as-data problem. Students move from an origin country to a destination country or city, and the attractiveness of a destination depends on economic conditions, institutional quality, distance, language, and student perceptions of daily life. This project asks:

> **What explains the size of Erasmus Higher Education mobility flows from one country to another, and how does the role of destination quality of life change over time?**

The starting point is a gravity-style view of student mobility. As with trade, migration, and commuting flows, Erasmus flows can be described as a function of origin conditions, destination pull factors, and bilateral frictions. The project focuses on three destination-pull variables: a broad prosperity control (GDP-per-capita gap), perceived quality of life, and institution quality, with distance and language affinity as frictions. The executed analysis is descriptive and exploratory; it does not claim causal effects.

The literature motivates this structure. Tinbergen's gravity model and the structural gravity framework of Anderson and van Wincoop explain why bilateral flows depend on both partner characteristics and frictions. Student-mobility studies such as Rodríguez González, Bustillo Mesanza, and Mariel (2011) and Beine, Noël, and Ragot (2014) apply this logic to international student flows, emphasising distance, cost, networks, and university quality. The text layer follows the "text as data" literature (Gentzkow, Kelly, and Taddy 2019; Grimmer and Stewart 2013): student discussions are treated as a noisy measurement of perceived quality of life, not as a representative survey. The PPML estimator follows Santos Silva and Tenreyro (2006) for count-flow gravity models with heteroskedasticity and zero flows.

This project addresses the course's project specification — a research question where text and spatial data are relevant, executed analysis with code and raw data, and a description of how the results answer the question — by combining official Erasmus mobility data, spatial aggregation and distance measures, an institution-quality layer from world university rankings, and a Reddit/OpenRouter/Pydantic extraction pipeline for student destination perceptions.

2. Data

The core data source is the official Erasmus+ statistical annex covering all mobility years 2014–2023:

`data/raw/annual-report-2024-statistical-annex.xlsx (sheet: HE-Learn Mobility)`

The pipeline filters the file to Higher Education learner study mobilities, keeps cross-border flows, and aggregates the result to origin-destination pairs (`sending_country`, `sending_code`, `receiving_country`, `receiving_code`, `flow_count`). The country-pair table contains 2,651 rows and 274,533 student flows. The largest corridors include Italy → Spain, Spain → Italy, Germany → Spain, France → Spain, and Germany →

France. These flows are the backbone of the spatial analysis. A multi-year city-level panel (2014–2023) is also built for the temporal extension.

Supporting datasets:

- **Economic indicators**: World Bank GDP per capita, Eurostat net earnings, and Eurostat price-level indicators.
- **Spatial controls**: great-circle distance and shared-language-family affinity between origin and destination capitals.
- **Institution quality**: QS World University Rankings 2023 (and 2014, 2017, 2018 retrieved via the Wayback Machine for the panel), geocoded with Nominatim and snapped to the nearest Erasmus destination city.
- **Cross-checks**: StudyPortals student ratings as a user-based institution-quality cross-check.
- **Temporal indicators**: OWID life satisfaction (Cantril ladder) and education-spending series; Eurostat Labour Cost Index (`lc_lci_r2_a`) and Labour Force Survey employment (`lfsa_egan2`) for the COVID-2020 sector shock.

The text layer uses Reddit `r/Erasmus` posts and comments. A structured extraction pipeline calls OpenRouter/DeepSeek and parses responses with a Pydantic schema; each extracted destination mention carries city, country, sentiment, sentiment score, themes (cost, housing, nightlife, academics, safety, bureaucracy, etc.), confidence, Reddit score, and permalink. The current extracted dataset contains **980 destination mentions across 56 countries and 211 cities: 360 positive, 363 neutral, and 257 negative mentions**. A 51-row stratified manual-validation sample (`data/processed/reddit_manual_validation.xlsx`, 17 per sentiment class) is provided alongside the submission so the LLM-vs-human agreement rate can be reported. The Reddit signal is interpreted as a qualitative perception proxy, not a representative measure of all Erasmus students.

3. Methods

Spatial flow pipeline. The Erasmus annex is filtered to the target population, reshaped and aggregated, then joined to country-level indicators. For each origin-destination pair the project computes monetary gaps `gdp_gap = receiver_gdp_pc − sender_gdp_pc`, `earnings_gap = receiver_net_earnings − sender_net_earnings`, and `price_level_gap = receiver_price_level − sender_price_level`. Flow-weighted destination summaries (e.g. `sum(flow × dest_gdp) / total_outflow`) describe whether a sending country tilts toward richer or poorer destinations.

Text pipeline. The Reddit extraction asks the model to identify destination mentions and classify each by sentiment and themes, with the schema mirroring the course's LLM information-extraction assignment: explicit JSON output, a Pydantic schema, and sentiment normalisation before aggregation. The Reddit signal is then cross-checked against an independent two-proxy quality-of-life construction: average per-city Erasmusu student ratings provide a parallel text-derived destination signal (Spearman $\rho \approx 0.21$ with the Reddit sentiment score), and both proxies enter the gravity model so the estimates do not lean on a single noisy text source.

Gravity design (step 3). The headline descriptive specification is a PPML gravity model of bilateral flows:

```
E[flow_ij] = exp(
    beta1 · gdp_gap_ij              # broad prosperity control
  + beta2 · quality_of_life_j       # Erasmusu rating (+ Reddit cross-check)
  + beta3 · institution_quality_j   # QS 2023 score, geocoded → city
  + gamma1 · log_distance_ij
  + gamma2 · same_language_ij
  + sending_country fixed effects
)
```

estimated with city-clustered robust standard errors. Salary gap and price-level gap are kept as robustness controls rather than headline terms, because they overlap strongly with the GDP gap (VIF ≈ 5–7 with each other) and make

the monetary block hard to interpret separately. Following Santos Silva and Tenreyro (2006), PPML is preferred to log-OLS because it handles zero flows and the multiplicative-error structure consistently.

Temporal extension. A city-year panel (2014–2023) is used to add (i) a time-varying β_2 via OWID life satisfaction — the Cantril ladder enters at the country-year level so identification is within-destination over time, with origin, destination, and year fixed effects; (ii) a time-varying β_3 via multi-year QS rankings (2014, 2017, 2018, 2023); and (iii) a 2020 COVID difference-in-differences at the field × NACE-sector cell level, estimated twice — once with Eurostat LCI wage growth (`covid_hit_pp`) and once with LFS employment growth (`covid_emp_hit_pp`) as the treatment-intensity measure.

Estimates are reported as exploratory associations, not causal effects.

4. Descriptive Results

The Erasmus flow structure is highly concentrated. Spain is central in the network — Italy → Spain and Germany → Spain are among the largest corridors — which supports the gravity intuition that destination popularity and institutional networks matter strongly.

The cross-sectional PPML output is directionally consistent with a destination-attractiveness story. Three findings stand out:

1. **Institution quality dominates.** The QS-based city score is positive and highly significant (β_3 = 0.012, p < 0.001), implying about **+28% per SD**. This should be read as academic + institutional attractiveness rather than pure teaching quality: top-ranked universities are also larger, more internationally visible, and embedded in denser Erasmus partnership networks.
2. **Quality of life adds independent signal.** The Erasmusu rating is positive and significant (β_2 = 0.159, p < 0.001) once institution quality is controlled for, equivalent to **+11% per SD**. The Reddit sentiment proxy is correlated with this measure (ρ ≈ 0.21), so the text layer enters as a robustness cross-check rather than the headline.
3. **Monetary gaps are weak headlines.** The GDP-per-capita gap is not significant in the headline cross-section ($\beta_1 = -3 \times 10^{-6}$, p ≈ 0.25). Salary and price-level gaps are correlated with GDP gap (VIF ≈ 5–7) and remain in the robustness column. Shared language-family affinity is positive and significant (β = 0.18, p ≈ 0.002).

The Reddit layer adds a qualitative view of destination perception. Spain, Italy, France, Poland, Portugal, and Germany dominate the extracted mention counts. Themes such as nightlife, academics, housing, bureaucracy, and cost pull in different directions for the same destination, so the layer is best read as evidence about *perceived* quality of life rather than as a precise ranking.

The **temporal extension** sharpens the QoL story. With panel year fixed effects, the OWID Cantril-ladder happiness score enters positively and significantly within destinations over time (β = 0.23, p ≈ 0.035; ≈ +19% per SD across 36 destinations, 2014–2023). This is the closest causal-adjacent evidence in the project because it identifies the QoL channel from within- destination changes, not just cross-country differences. By contrast, the time-varying QS score is null (p ≈ 0.57), consistent with rank stickiness — once a city has a top university it tends to keep it.

The **COVID-2020 sector DiD** is informative for what it does *not* find. With the Eurostat LCI wage-growth hit as treatment intensity, the estimate is small and weakly negative (p ≈ 0.10); with the cleaner LFS employment-growth hit it collapses to zero (p ≈ 0.95). The honest reading is that COVID's mobility shock was **lockdown-wide, not field-selective**: Erasmus flows in 2020 fell across the board, not preferentially in the sectors that took the bigger wage or employment hit. The wage-based version partly picked up lockdown intensity because LCI is contaminated by 2020 wage-subsidy schemes (Dutch NOW, German Kurzarbeit), which the LFS replication confirms.

5. Interpretation, Limitations, and Conclusion

The project answers the research question on two fronts. First, the spatial flow analysis shows where students move and how destination profiles differ by sending country. Second, the text layer adds a student-perception variable not available in official Erasmus data. Together they show how a gravity model of Erasmus mobility can combine mobility flows, economic indicators, spatial controls, institution quality, and student-generated text.

The clearest result is not a causal coefficient but a descriptive pattern: destination attractiveness is multidimensional. High-flow destinations combine strong institutions, favourable perceived living conditions, and familiar language environments. Monetary gaps are difficult to separate because they capture overlapping rich-destination dimensions. The quality-of-life signal weakens under stricter fixed effects, suggesting much of it lives across countries rather than within countries across cities — plausible because Erasmus mobility is constrained by bilateral university agreements, not only by free student choice. The OWID time-varying panel is the strongest piece of quasi-experimental evidence because it identifies QoL within destination over time.

Limitations. Reddit users are not representative of all Erasmus students, and per-country mention counts are small. The manual-validation xlsx provides the required hand-labelled cross-check on a stratified sample of 51 mentions, but the Reddit layer should still be read as a perception proxy. The institution-quality measure relies on Nominatim geocoding plus nearest-city snapping, which introduces assignment noise, and it cannot separate academic quality from institutional scale: a city with a top-ranked university tends to have larger campuses, more exchange seats, and more bilateral agreements. Without a causal research design, all regression estimates are exploratory associations.

Contribution. The project builds a reproducible descriptive pipeline for Erasmus Higher Education mobility, combines spatial flow analysis with economic indicators and an institution-quality layer, and adds a text-derived perception layer from Reddit. It documents that institution quality, perceived quality of life, and language affinity are positively associated with Erasmus destination flows conditional on broad prosperity and distance, that the quality-of-life association is robust within destinations over time, and that the COVID-2020 mobility shock was lockdown-wide rather than field-selective.

References

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